

Number System

	Base (radix)	digit
Decimal	10	0, 1, ..., 9
Binary	2	0, 1
Octal	8	0, 1, ..., 7
Hexadecimal	16	0, 1, ..., F

(652)_x

→ LSB ⇒ Least significant binary
 → MSB ⇒ More significant binary

4 bit - 1 nibble
 8 bit - 1 byte
 16 bit - 1 word - 2 byte - 4 nibble

Base 4 → 2-2 grouping

Base 8 → 3-3 — 11 —

Base 16 → 4-4 ~~11~~ — 11 —

Digit	Base-4	Base-8	Base-16
0	00	000	0000
1	01	001	0001
2	10	010	0010
3	11	011	0011
4		100	0100
5		101	0101
6		110	0110
7		111	0111

8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Ex: $(101010)_2 \Rightarrow (?)_4 \Rightarrow (?)_8 \Rightarrow (?)_{16}$

$$\begin{array}{c} 101010 \\ \hline 2 \quad 2 \quad 2 \end{array} \Rightarrow (222)_4$$

$$\begin{array}{c} 101010 \\ \hline 5 \quad 2 \end{array} \Rightarrow (52)_8$$

$$\begin{array}{c} 00101010 \\ \hline 2 \quad A \end{array} \Rightarrow (2A)_{16}$$

Ex: $(10110101)_2$

$$10110101 \Rightarrow (2311)_4$$

$$10110101 \Rightarrow (265)_8$$

$$10110101 \Rightarrow (B5)_{16}$$

Ex: $(ABCD)_{16} =$

$$(1010101111001101)_2$$

$$1010101111001101 \Rightarrow (2223303D)_4$$

$$1010101111001101 \Rightarrow (125715)_8$$

Ex: $(101011.10101)_2$

$$101011.10101 \Rightarrow (53.52)_8$$

$$101011.10101 \Rightarrow (213.A8)_{16}$$

Decimal to any base system:

① Decimal to binary:

Ex: $(45)_{10} = (?)_2$

2	45	
2	22	1
2	11	0
2	5	1
2	2	1
	1	0

$$\Rightarrow (101101)_2$$

Ex: $(45.625)_{10}$

$$.625 \times 2 = 1.250$$

$$.25 \times 2 = 0.500$$

$$.50 \times 2 = 1.00$$

1
0
1
↓

$$= (101101.101)_2$$

Decimal to octal:

Ex: $(378)_{10}$

8	378	
8	47	2
	5	7

$$\Rightarrow (572)_8$$

Ex: $(378.93)_{10}$

$$.93 \times 8 = 7.44$$

$$.44 \times 8 = 3.52$$

$$.52 \times 8 = 4.16$$

$$.16 \times 8 = 1.28$$

7
3
4
1
↓

$$(572.7341)_8$$

Ex: $(172.878)_{10}$

8	172	
8	21	4
	2	5

$$\begin{aligned}
 & \cdot 878 \times 8 = 7.024 & 7 \\
 & \cdot 024 \times 8 = 0.192 & 0 \\
 & \cdot 192 \times 8 = 1.536 & 1 \\
 & \cdot 536 \times 8 = 4.288 & 4
 \end{aligned}$$

$$(254.7014)_8$$

③ Decimal to hexadecimal ÷

Ex: $(2598.675)_{10}$

16	2598	
16	162	6
	10	2

$$\begin{aligned}
 & \cdot 675 \times 16 = 10.8 \\
 & \cdot 8 \times 16 = 12.8 \\
 & \cdot 8 \times 16 = 12.8
 \end{aligned}$$

$$(A26.ACC)_{16}$$

Anybase to decimal ÷

① Binary to decimal ÷

Ex: $(101101)_2$

$$\begin{aligned}
 & = 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\
 & = 45
 \end{aligned}$$

Ex: $(101101.101)_2$

$$= 45.1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 45.625$$

② Octal to decimal:

Ex: $(4057.06)_8$

$$= 4 \times 8^3 + 0 \times 8^2 + 5 \times 8^1 + 7 \times 8^0 + 0 \times 8^{-1} + 6 \times 8^{-2}$$

$$= 2095.09375$$

③ Hexadecimal to decimal:

Ex: $(5C7)_{16}$

$$= 5 \times 16^2 + 12 \times 16^1 + 7 \times 16^0$$

$$= 1479$$

Representation of number:

- Unsigned
- Signed

Ex: $(1) + 8 = 01000$

$$- 8 = 11000$$

$(2) + 7 = 0111$

$$- 7 = 1111$$

There are 3 methods to represent unsigned no. :-

- (i) Sign magnitude form
- (2) 1's comp. ——— 11 ———
- (3) 2's ——— 11 ———

(i) Sign magnitude form :-

Ex: $+4 = \begin{array}{|c|c|c|c|} \hline 0 & 1 & 0 & 0 \\ \hline \end{array} = 000100 = 00000100$
 $-4 = \begin{array}{|c|c|c|c|} \hline 1 & 1 & 0 & 0 \\ \hline \end{array} = 100100 = 10000100$
 Sign ← 4 bit magnitude 8 bit

$+5 = 0101 = 00000101$

$-5 = 1101 = 10000101$

Range of sign & magnitude form :-

Unsigned	Decimal	Signed	Decimal
00	+0	00	+0
01	+1	01	+1
10	+2	10	-0
11	+3	11	-1

Range = -1 to $+1$

Range of a number :-

Range of a given no. having 'n' bit

in sign magnitude form is :-

$= -(2^{n-1} - 1) \text{ to } +(2^{n-1} - 1)$

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for $n=2$

$$= -(2^2 - 1) \text{ to } +(2^2 - 1) \\ = -1 \text{ to } +1 \quad \Rightarrow -1, 0, +1$$

for $n=3$

$$= -(2^3 - 1) \text{ to } +(2^3 - 1) \\ = -(4 - 1) \text{ to } +(4 - 1) \\ = -3 \text{ to } +3$$

Signed Decimal

000 +0

001 +1

010 +2

011 +3

100 -0

101 -1

110 -2

111 -3

Ex: 0101 = +5
1101 = -5

for $n=4$

$$= -(2^4 - 1) \text{ to } +(2^4 - 1) \\ = -(8 - 1) \text{ to } +(8 - 1) \\ = -7 \text{ to } 7$$

Q. $x = -4 = 1100$

$y = +3 = +0011$

$-1 = 1111$ (-7) wrong

$x = -4 = 1100$

$y = -3 = +1011$

110111

(+7) wrong

Disadvantage:

- 2 diff. representation of zeroes (-0, +0)
- Calculation is not easy in this form.

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(x-1)'s or 1's complement ÷

$$+x = +x$$

$$-x = \overline{+x}$$

$$+4 = 0100$$

$$-4 = \overline{+4} = \overline{0100} \Rightarrow 1011 \text{ (-ve)}$$

Note:

In 1's complement form or 2's complement or sign magnitude form. Positive number apply zero & neg. no. apply one.

$$+9 = 01001$$

$$-9 = \overline{01001} \Rightarrow 10110$$

$$8 \text{ bit } \Rightarrow +9 = 00001001$$

$$-9 = 11110110$$

Q. Given data = 10010 is in 1's complement form. So corresponding value will be?

$$1's = \overline{10010} = 01101 = -13$$

Q. n = 2 bit

Binary	Decimal
00	+0
01	+1
10	-1
11	-0

Range $\Rightarrow -1, -0, +0, +1$

n bit

Range $\Rightarrow -(2^{n-1}-1)$ to $+(2^{n-1}-1)$

For $n=3$ bit

Binary	Decimal
000	+0
001	+1
010	+2
011	+3
100	-3
101	-2
110	-1
111	-0

Q. $x = -4$, $y = +3$

$$x = -4 = \overline{0100} = 1011$$

$$y = +3 = \overline{0011}$$

$$\begin{array}{r} 1011 \\ + 0011 \\ \hline \end{array}$$

$$\underline{1110}$$

$$\Rightarrow \overline{1110} = 0001 (-1)$$

$$(x-1)'s \text{ complement} = (x^n - 1) - N$$

$x = \text{base}$

$n = \text{no. of digit}$

$N = \text{Number}$

Ex: ① $N = 101$

$$1's \text{ complement} = (2^3 - 1) - 101$$

$$= (1000 - 1) - 101 \Rightarrow 111 - 101 \Rightarrow 010$$

$$\begin{array}{r} 1010.010 \\ - 111.111 \\ \hline 0010.011 \end{array}$$

$$\left[\begin{array}{l} 0-0=0, \quad 1-0=1 \\ 1-1=0, \quad 0-1=1 \end{array} \right]$$

② Find out the 1's comp. of $(10110)_2$

$$(2^5 - 1) - 10110$$

$$(100000 - 1) - 10110$$

$$= 11111 - 10110$$

$$= 01001$$

$$\begin{aligned}
 \textcircled{3} \text{ Find } (1101.1011)_2 \\
 &= (2^8 - 1) - 11011011 \\
 &= (100000000 - 1) - 11011011 \\
 &= 11111111 - 11011011 \Rightarrow 0010.0100
 \end{aligned}$$

9's Complement:

$$(10^n - 1) - N$$

$$\textcircled{1} (25)_{10}$$

$$\begin{aligned}
 &(10^2 - 1) - 25 \\
 &= (100 - 1) - 25 \\
 &= 99 - 25 \Rightarrow 74
 \end{aligned}$$

$$\textcircled{2} (413062)_{10}$$

$$\begin{aligned}
 &= (10^6 - 1) - 413062 \\
 &= (1000000 - 1) - 413062 \\
 &= 999999 - 413062 \\
 &= 586937
 \end{aligned}$$

7's complement:

$$(8^n - 1) - N$$

$$\textcircled{1} (35)_8$$

$$(8^2 - 1) - 35$$

$$= \cancel{(64 - 1) - 35} \quad \cancel{(77 - 1) - 35}$$

$$= \cancel{63 - 35} \quad \cancel{76 - 35} = 42$$

$$\textcircled{2} (413.42)_8$$

$$= (8^5 - 1) - 413.42$$

$$= 32767 - 413.42$$

$$= 77777 - 413.42$$

$$= 364.35$$

15's complement:

$$(16^n - 1) - N$$

$$\textcircled{1} (A13.21)_{16}$$

$$= (16^5 - 1) - A13.21$$

$$= FFFFF - A13.21$$

$$= 5EC.9E$$

Radix complement (e's comp.):

The e's comp. of a no. N , consisting n -bit which has e base

$$= e^n - N$$

This can be also written as $[(e^n - 1) - N] + 1$

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① Find out the 2's comp. of $(1011010)_2$

$$\begin{aligned} 2's &= 2^m - N \\ &= 2^7 - 1011010 \\ &= 128 - 1011010 \\ &= 10000000 - 1011010 \\ &= 0100110 \end{aligned}$$

② $(110101)_2$

$$\begin{aligned} 2's &= 2^6 - 110101 \\ &= 64 - 110101 \\ &= 1000000 - 110101 \\ &= 001011 \end{aligned}$$

③ A number is expressed in 2's comp. as 10011. Find its binary value.

$$\begin{aligned} &= 2^m - N \\ &= 2^5 - 10011 \\ &= 100000 - 10011 \\ &= 01101 \end{aligned}$$

10's complement $\div$ $= 10^n - N$

① Find out 10's comp. of $(132410)_{10}$

$$= 10^6 - 132410$$

$$= 1000000 - 132410$$

$$= 867590$$

8's complement:

$$= 8^n - N$$

$$\textcircled{1} (1374)_8$$

$$= 8^4 - 1374$$

$$= \cancel{7777} - \cancel{1374} = 10000 - 1374$$

$$= \cancel{6403} = 86404$$

② 16's complement:

$$= 16^n - N$$

$$\textcircled{1} (1B0.19)_{16}$$

$$= 16^5 - 1B019$$

$$= 1048576 - 1B019$$

$$= 1000000 - 1B019$$

$$= \cancel{FEE7} \boxed{E4F.E7}$$

Subtraction using complements:

Sub. using 1's comp.:

Step-1:-

Convert the no. to be sub. to its 1's comp. form.

Step-2:-

Perform the addition.

Step-3:-

If the carry is 1 (ese) (EAC) then add it to the result obtained in step 2. (For sub. of smaller no. to larger no.)

Step-4:-

If the carry is 0 result obtained in (-)ive & in its first 1's comp. form. So do the 1's comp. again to get the answer. (For sub. of larger no. to smaller no.)

Case-1 Sub. of smaller no. from larger no.

① Sub. $(1011)_2$ from $(1101)_2$

1's comp. of $1011 \rightarrow 0100$

1101

$+ 0100$

$\hline 10001$

$+ 1$

$\hline 0010$

$= (2)_{10}$

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2) Perform $(1100)_2 - (0101)_2$

1's comp. of $\rightarrow 1100 \rightarrow 0011$ $\rightarrow 1010$

$$\begin{array}{r}
 \cancel{0101} \\
 + \cancel{0011} \\
 \hline
 \cancel{0000}
 \end{array}
 \quad
 \begin{array}{r}
 1100 \\
 1010 \\
 \hline
 10110 \\
 + 1 \\
 \hline
 0111 \Rightarrow (7)_{10}
 \end{array}$$

3) Sub. $(011010)_2$ from $(1111101)_2$

1's comp. of $1111101 \rightarrow 0000010$

1's comp. of $011010 \rightarrow 1000101$

$$\begin{array}{r}
 \cancel{1111101} \\
 + \cancel{0000101} \\
 \hline
 \cancel{1111101} \\
 + \cancel{0000101} \\
 \hline
 1000011
 \end{array}
 \quad
 \begin{array}{r}
 1111101 \\
 111010 \\
 \hline
 1000011
 \end{array}$$

$= (67)_{10}$

4) $(11010100 - 10101001)_2$

1's comp. of $10101001 \rightarrow 01010110$

$$\begin{array}{r}
 11010100 \\
 + 01010110 \\
 \hline
 1000101010 \\
 + 1 \\
 \hline
 00101011
 \end{array}
 \quad
 = (43)_{10}$$

Case - ②

Sub. of larger number from smaller no.

① $(10011)_2 - (11010)_2$

1's comp. of $11010 \rightarrow 00101$

$$\begin{array}{r} 10011 \\ + 00101 \\ \hline 11000 \\ \hline \end{array}$$

Real ans = $11000 = 00111 \Rightarrow (7)_{10}$

Ans $(-7)_{10}$

② $(1101)_2 - (1110)_2$

1's comp. of $1110 \rightarrow 0001$

$$\begin{array}{r} 1101 \\ + 0001 \\ \hline 1110 \\ \hline \end{array}$$

Real = $1110 \Rightarrow 0001 \Rightarrow (1)_{10}$

Ans $(-1)_{10}$

no Binary sub. using 2's comp. +

Step-① ÷

Find 2's comp. of a no. to be subtracted.

Step-② ÷

Perform the addition.

Step-③ ÷

If the final carry is generated then the result is (+)ive & in its true form (Sub. for smaller from larger)

Step-④ ÷

If final carry is not produced then the result is (-)ive & in its 2's comp. form.

① $(101101)_2 - (100110)_2$ using 2's comp.

2's comp. of $100110 \rightarrow 011001 + 1 \Rightarrow 011010$

$$\begin{array}{r} 101101 \\ + 011010 \\ \hline 1000100 \end{array}$$

$$\begin{array}{r} 101101 \\ + 011010 \\ \hline 1100011 \end{array}$$

↑
discard

$$(000111) \Rightarrow (7)_{10}$$

② $(10110101)_2 - (10101010)_2$

2's comp. of $10101010 \Rightarrow 01010101 + 1$
 $= 01010110$

$$\begin{array}{r} 10110101 \\ + 01010110 \\ \hline 1100001011 \end{array}$$

(00001011)

Case-2 Sub. of larger no. from smaller no.

① $(110101 - 111010)_2$

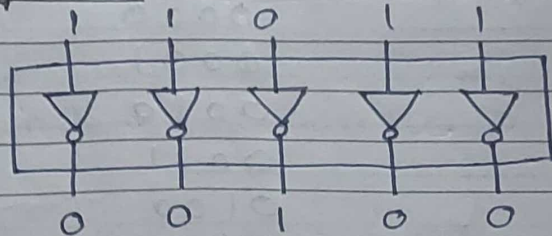
2's comp. of $111010 \rightarrow 000101 + 1 \Rightarrow 000110$

$$\begin{array}{r} 110101 \\ + 000110 \\ \hline 111011 \end{array}$$

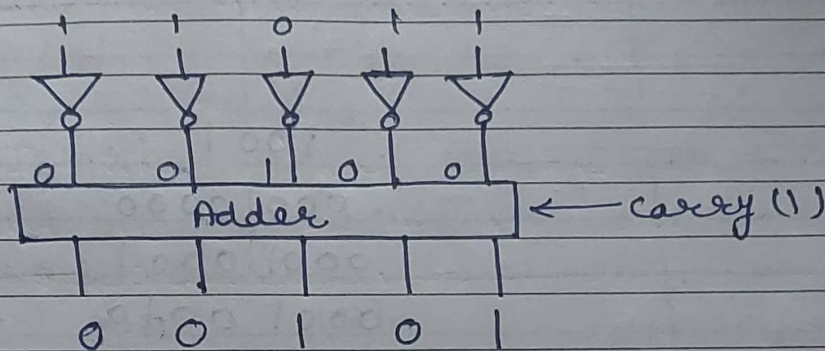
2's comp. of $111011 \rightarrow 000100 + 1 \Rightarrow 000101$
 $= 000101 \neq 000100 \Rightarrow (-5)_{10}$

Comparison b/w 1's and 2's comp. :

1's comp. $\Rightarrow$



2's comp. $\Rightarrow$



BCD (Binary coded decimal) $\div$

BCD is a way to express a decimal digit with 4 bit binary code.

• It represents decimal digits from 0-9 same as binary.

• 8421 is type of BCD which means that each decimal bit in any no. is represented by 4 bit binary equivalent.

Decimal	Binary	BCD
0	0	0000
1	01	0001
2	10	0010
3	11	0011
4	100	0100
5	101	0101
6		
7		
8		
9		1001
10		00010000
11		00010001
12		00010010
13		00010011
14		00010100
15		00010101

① 84 in BCD and 246 in BCD

$$84 = 10000100$$

$$246 = 001001000110$$

BCD addition:

Add 2 BCD no. like normal binary addition.

(a) If ^{Sum} ~~some~~ ≤ 9 , if final carry = 0 then answer

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is correct.

(b) If ~~some~~ ^{sum} ≤ 9 , and final carry = 1, so answer is incorrect. So add 6 (0110).

(c) If ~~sum~~ ^{sum} > 9 & final carry = 0, so answer is incorrect. (Add 6)

① $(2)_{10} + (6)_{10}$
 $(0010)_{10} + (0110)_{10}$

$$\begin{array}{r} 0010 \\ + 0110 \\ \hline 1000 \end{array} \quad (8)_{10}$$

② $(3)_{10} + (7)_{10}$
 $(0011)_{10} + (0111)_{10}$

$$\begin{array}{r} 0011 \\ + 0111 \\ \hline 1010 \end{array} \quad (10)_{10} \Rightarrow \text{invalid}$$

Sum > 9

$$\begin{array}{r} 1010 \\ + 0110 \\ \hline 00010000 \end{array}$$

② $(8)_{10} + (9)_{10}$

$$\begin{array}{r} 1000 \\ + 1001 \\ \hline 10001 \end{array}$$

$$\begin{array}{r} 10001 \\ 0110 \\ \hline 00010111 \end{array} \quad (17)_{10}$$

④ $(57)_{10} + (26)_{10}$

$$(01010111)_{10} + (00100110)_{10}$$

$$\begin{array}{r} 01010111 \\ + 00100110 \\ \hline 01111101 \end{array}$$

7 13

$$\begin{array}{r} 01111010 \\ + 0110 \\ \hline \cancel{01111010} \\ 10000011 \end{array}$$

8 3

$(83)_{10}$

BCD Subtraction:

Algorithm of BCD

Decode result	Sign of total result	
	(+) EAC = 1	(-) EAC = 0
	Transfer true result of adder 1 to adder 2	Transfer 1's comp of adder 1 to 2
Cn = 1	0000 added in 2	1010 added in 2
Cn = 0	1010 added in 2	0000 added in 2

① $(546)_{10} - (429)_{10}$

$(010101000110)_{10} - (010000101001)_{10}$

$$\begin{array}{r} 010101000110 \\ 010000101001 \\ \hline \end{array}$$

1's

101111010110

$\Rightarrow 010101000110$

$+ 101111010110$

$\Rightarrow 000000000100$
→ EAC

$$\begin{array}{r} 000000001100 \\ + 1111101101 \\ \hline 000100011101 \\ 000000001010 \\ \hline 000100010111 \\ \hline \end{array}$$

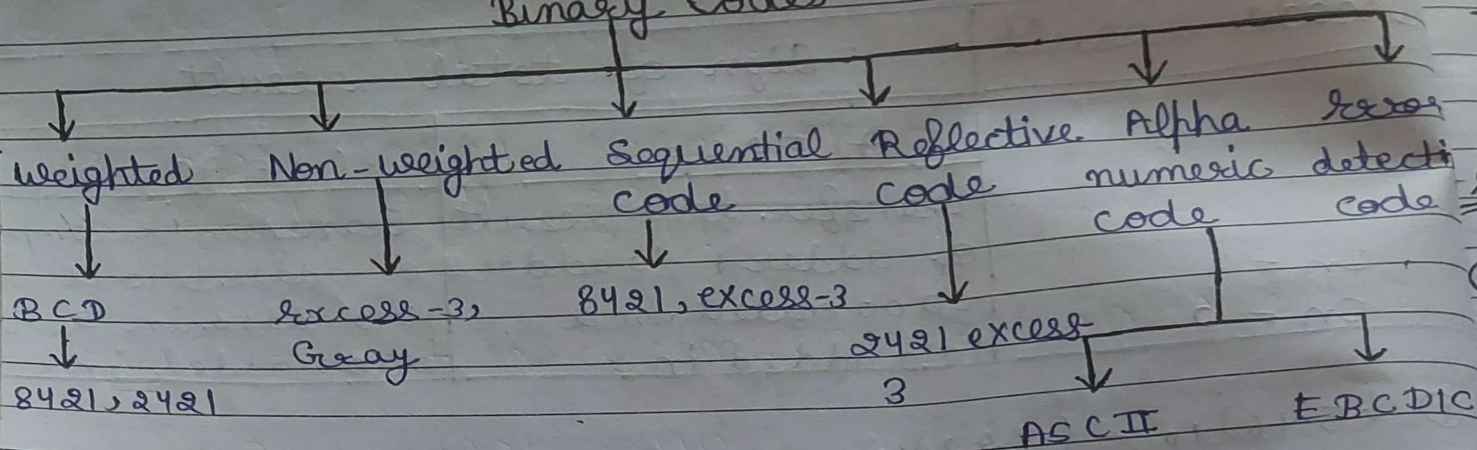
1 1 ↓ discard 7

$\Rightarrow (117)_{10}$

BCD = $(000100010111)_{10}$

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Binary Codes



2421 ÷

Decimal	8421	2421
0	0000	0000
1	0001	0001
2	0010	0010
3	0011	0011
4	0100	0100
5	0101	1011
6	0110	1100
7	0111	1101
8	1000	1110
9	1001	1111

Reflective
Sequential X

Non-weighted ÷

• Excess - 3 code :-

As name specifies it is the representation of decimal no. as BCD.

Ex: $(5)_{10} \xrightarrow{\text{excess-3}} 0101 + 0011 = 1000$

① Rewrite down the excess-3 code of $(25)_{10}$

$(25)_{10} \xrightarrow{\text{excess-3}} 000100101 + 0011 \xrightarrow{+0011} \cancel{00101000} \Rightarrow 01011000 (58)$

Decimal	8421	2421
0	0000	0001
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

① Convert 6067 in excess-3 code.

$$\begin{array}{cccc} 0110 & 0000 & \cancel{0011} & 0111 \\ 0011 & 0011 & 0011 & 0011 \\ \hline 1001 & 0011 & 1001 & 1010 \end{array}$$

② convert excess-3 code in decimal.
(0011 0101 1010 0100).

$$\begin{array}{r}
 0011 \quad 0101 \quad 1010 \quad 0100 \\
 - 0011 \quad 0011 \quad 0011 \quad 0011 \\
 \hline
 0000 \quad 0010 \quad 0111 \quad 0001
 \end{array}$$

Ques. Convert the decimal no 396 in excess-3.

$$\begin{array}{r}
 0011 \quad 1001 \quad 0110 \\
 + 0011 \quad 0011 \quad 0011 \\
 \hline
 0110 \quad 1100 \quad 1001 \\
 \hline
 6 \quad 12 \quad 9
 \end{array}$$

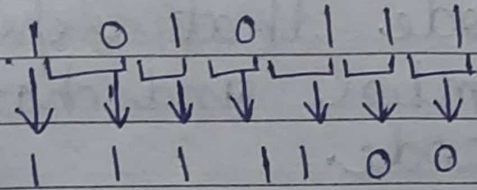
Gray code:

It is also known as unit distance or minimum change code b/c each ~~code~~ code changes only by 1 bit from the preceding code.

Decimal	8421	Gray code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101

7	0 1 1 1	0 1 0 0
8	1 0 0 0	1 1 0 0
9	1 0 0 1	1 1 0 1
10		1 1 1 1
11		1 1 1 0
12		1 0 1 0
13		1 0 1 1
14		1 0 0 1

Ex: $(1010111)_2 \rightarrow \text{gray}$



Ex: 0 1 0 1 0 1 0 1
↓
0 1 1 1 1 1 1 1

Ex: $(1110011)_2$

1 0 0 1 0 1 0

Conversion of gray code to binary :-

Q. $\begin{array}{cccccc} 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ \downarrow & \swarrow & \downarrow & \swarrow & \downarrow & \swarrow & \downarrow \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{array}$

Q. $\begin{array}{cccccc} 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ \downarrow & \swarrow & \downarrow & \swarrow & \downarrow & \swarrow & \downarrow & \swarrow \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{array}$

Alpha numeric code :-

The code that represent no., alphabet, special symbol and charac. is called alpha numeric code.

It include

- (a) 26 lower. keys alphabet
- (b) 26 upper " "
- (c) 10 numeric digits
- (d) about 25 special character.

(a) ASCII :-

In ASCII code ^{each} charac. is represent by 7 bit code. So, total no. of combination will be $2^7 = 128$.

It stands for american standard code for information interchange.

Q. HI
100 1000 100 100

Q(b) EBCDIC code ÷

It stands for extended binary coded decimal information interchange code.

Total no. of combination = $2^8 = 256$

Q. D
1100 0100

Q. C
1100 0011

Error detecting code ÷

Since there may be a quiet possibility of error due to noise during transmitting the code, from transmitter to receiver in digital system.

- Since digital system must be accurate to the digits. So, error can become a serious problem.

• To detect the error several schemes have been designed. So, that transmitted information is distorted.

• One of the most popular method is parity
 ⇒ Even parity
 ⇒ Odd parity

Information	Even parity	odd parity
A B C	A B C P	A B C P
0 0 0	0 0 0 0	0 0 0 1
0 0 1	0 0 1 1	0 0 1 0
0 0 1	0 0 1 1	0 0 1 0
0 1 0	0 1 0 0	0 1 0 1
0 1 1	0 1 1 1	0 1 1 0
1 0 0	1 0 0 1	1 0 0 0
1 0 1	1 0 1 0	1 0 1 1
1 1 0	1 1 0 0	1 1 0 1
1 1 1	1 1 1 1	1 1 1 0

If the system is working odd parity and if receiver is getting even number then the data will be rejected and vice versa.

Limitation:-

This method ~~decates~~ detects only one error in a request sample at a time. • If 2

errors occurs in a sample at a time, then it does not detect the error and then we used the concept of check sum.

checksum:

In this each word is added to the previous word then we get total sum. after this one's complement of the total sum will be our checksum. Now it is transmitted along with the data to the receiver side and at the receiving end, the total sum of receiver is added to the checksum of sender side and then if the 1's comp. of resultant is 0 then there is no error.

Ex: Suppose that the sender one's to send 4 frames each of 8 bits, where the frames are 11001100, 10101010, 11110000, 11000011,

At the sender side:

Frame-1	11001100
Frame-2	+ 10101010
	<u>1'01110110</u>

$$\begin{array}{r} \text{Frame-3} + 11110000 \\ - 1100110 \\ \hline \end{array}$$

$$01110110$$

$$01110111$$

$$\text{Frame-3} + 11110000$$

$$(1) 01100111$$

$$\rightarrow 1$$

$$01101000$$

$$\begin{array}{r} \text{Frame-4} + 11000011 \\ - 10010101 \\ \hline \end{array}$$

$$+ 1$$

$$00101100$$

$$\begin{array}{r} \text{Checksum } 00101100 \\ \Rightarrow 11010011 \end{array}$$

At the receiver end

$$\text{Frame-1 } 11001100$$

$$\text{Frame-2 } 10101010$$

$$10111110$$

$$10111011$$

$$\rightarrow 1$$

$$01110111$$

$$\text{Frame-3} + 11110000$$

$$10110011$$

$$\begin{array}{r} 01100111 \\ + 1 \\ \hline 01101000 \end{array}$$

$$\begin{array}{r} \text{Frame-4 } 11000011 \\ + 10010101 \\ \hline 00101100 \end{array}$$

checksum = 11010011

$$\begin{array}{r} \text{Total sum} = 00101100 \\ + 11010011 \\ \hline 11111111 \end{array}$$

$\Rightarrow 1's = 00000000 = 0$ (No error)

So, receiver will accept the data.

② Suppose that the sender is to send 4 frames each of 8 bits, where the frames are ~~10110011~~ 10110011, 10101011, 10101101, 11010101

At the sender side

$$\begin{array}{r} \text{Frame-1 } 10110011 \\ \text{Frame-2 } + 10101011 \\ \hline 101011110 \\ \hline 01011111 \end{array}$$

$$\begin{array}{r}
 01011110 \\
 +1 \\
 \hline
 01011111 \\
 \text{frame 3} - 01011010 \\
 \hline
 10111001 \\
 +1 \\
 \hline
 01111010 \\
 \text{frame 4} - 11010101 \\
 \hline
 10000111 \\
 +1 \\
 \hline
 00010000
 \end{array}$$

$$\text{checksum} = 11101111$$

Receiver side

$$\begin{array}{r}
 \text{frame 1} - 10110011 \\
 \text{frame 2} - 10101011 \\
 \hline
 10101110 \\
 \hline
 01011111 \\
 \text{frame 3} - 01011010 \\
 \hline
 10111001 \\
 \text{frame 4} - 11010101 \\
 \hline
 110001110 \\
 \hline
 10001111
 \end{array}$$

$$\begin{array}{r}
 01011111 \\
 \text{frame 3} - 01011010 \\
 \hline
 10111001 \\
 \text{frame 4} - 11010101 \\
 \hline
 10001110 \\
 \hline
 10001111
 \end{array}$$

$$\begin{array}{l}
 \text{checksum} = 10001111 \\
 = 01110000
 \end{array}$$

$$\text{checksum} = 01110000$$

$$\text{total sum} = 10001111$$

$$11111111$$

$$00000000$$

Error correcting codes :

Hamming code :

It is a technique developed by R. W. Hamming to detect the error msg. The ~~sp~~ parity bits in hamming code the source ~~in~~ encodes the msg by adding parity bits. In the msg. these parity bits are inserted & generated at certain position in the msg to accomplish the error detection & correction process.

Power of parity bits = $2^n = n = [0, 1, 2, \dots, n]$

$$2^0 \rightarrow 1 \quad (P_1)$$

$$2^1 \rightarrow 2 \quad (P_2)$$

$$2^2 \rightarrow 4 \quad (P_4)$$

$$2^3 \rightarrow 8 \quad (P_8)$$

$$\Rightarrow 2^p \geq m + p + 1 \rightarrow \text{no. of parity bits}$$

$\downarrow$
 no. of msg bits

msg = 1001
 $m = 4$

$$2^0 \geq 4 + 0 + 1$$

$$1 \geq 5$$

$$p = 3 \quad = \quad 2^3 \geq 4 + 3 + 1$$

$$8 \geq 8$$

$$p = 3$$

$$m + p \Rightarrow 4 + 3 = 7 \text{ (Hamming code)}$$

7	6	5	4	3	2	1
1	0	0		1		
D_7	D_6	D_5	P_4	D_3	P_2	P_1

	P_4	P_2	P_1
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1

$$P_1 = D_3 D_5 D_7$$

$$P_2 = D_3 D_6 D_7$$

$$P_4 = D_5 D_6 D_7$$

$$4 \quad 1 \ 0 \ 0$$

$$P_1 = D_3 D_5 D_7$$

$$5 \quad 1 \ 0 \ 1$$

$$1 \ 0 \ 1 \quad P_1 = 0$$

$$6 \quad 1 \ 1 \ 0$$

$$7 \quad 1 \ 1 \ 1$$

$$P_2 = D_3 D_6 D_7$$

$$1 \ 0 \ 1 \quad P_2 = 0$$

$$P_4 = D_5 D_6 D_7$$

$$0 \ 0 \ 1 \quad P_4 = 1$$

1	0	0	1	1	0	0
---	---	---	---	---	---	---

Hamming code = 1001100

$\rightarrow$ 1101100

1	1	0	1	1	0	0
---	---	---	---	---	---	---

$$P_1 = D_3 \ D_5 \ D_7$$

$$\downarrow$$

$$0 \quad 1 \quad 0 \quad 1$$

$$P_1 = 0 \quad \checkmark$$

$$P_2 = D_3 \ D_6 \ D_7$$

$$0 \quad 1 \quad 1 \quad 1$$

$$P_2 = 1 \quad X$$

$$P_4 = D_5 \ D_6 \ D_7$$

$$1 \quad 0 \quad 1 \quad 1$$

$$P_4 = 1 \quad X$$

$$P_4 \ P_2 \ P_1 = 110 = (6)_{10}$$

There is a error at 6th position.

So, the corrected hamming code = 1001100

⇒ Data bits 1011 is to be transmitted
construct the even parity 7 bits hamming code.

$$m = 4$$

$$2^p \geq m + p + 1$$

$$p = 0$$

$$2^0 \geq 4 + 0 + 1$$

$$1 \geq 5$$

$$p = 3$$

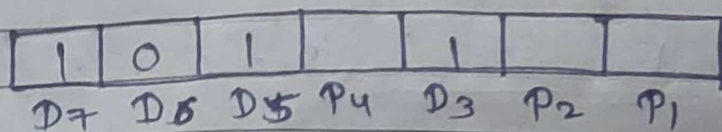
$$2^3 \geq 4 + 3 + 1$$

$$8 \geq 8$$

$$p = 3$$

$$m + p = 4 + 3$$

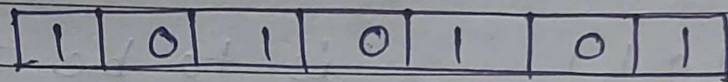
$$= 7$$



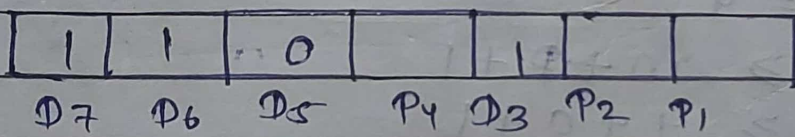
P ₄	P ₂	P ₁
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

$P_1 = D_3 \oplus D_5 \oplus D_7$
 $P_2 = D_4 \oplus D_6 \oplus D_7$
 $P_4 = D_5 \oplus D_6 \oplus D_7$

$P_1 = 1$
 $P_2 = 0$
 $P_4 = 0$



Q. Make even parity 7 bit hamming code for 1101.



$P_1 = D_3 \oplus D_5 \oplus D_7$
 $1 \oplus 0 \oplus 1 \Rightarrow P_1 = 0$

$P_2 = D_4 \oplus D_6 \oplus D_7$
 $1 \oplus 1 \oplus 1 \Rightarrow P_2 = 1$

$P_4 = D_5 \oplus D_6 \oplus D_7$
 $= 0 \oplus 1 \oplus 1 \Rightarrow P_4 = 0$

1 1 0 0 1 1 0

Q. A 7 bit hamming code coming out of transmission line is 1000010. What was the code transmitted assume even parity.

1 0 0 0 0 1 0
D₇ D₆ D₅ P₄ D₃ P₂ P₁

$$P_1 = D_3 D_5 D_7 \\ 0 \quad 0 \quad 1 \Rightarrow 1$$

$$P_2 = D_3 D_6 D_7 \\ = 0 \quad 0 \quad 1 \Rightarrow 1$$

$$P_4 = D_5 D_6 D_7 \\ = 0 \quad 0 \quad 1 \Rightarrow 1$$

There is a error at 7th position. $\Rightarrow (111) \Rightarrow (7)_{10}$

0 0 0 0 0 1 0
D₇ D₆ D₅ P₄ D₃ P₂ P₁

Q. Given 8 bit data . 10111001 . Generate the 12 bit compasit word for ham. code that detects and ~~corresponde~~ correct single bit error using even parity at the receiving end , it can be received as 101101001110.

1	0	1	1	1	0	0	1
---	---	---	---	---	---	---	---

Received H.C. at receiver side = 101101001110
 $m=8$

1	0	1	1	0	1	0	0	1	1	0	0
D_{12}	D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1

$$P_1 = D_3 \ D_5 \ D_7 \ D_9 \ D_{11} \\ 1 \ 0 \ 1 \ 1 \ 0 \Rightarrow 01$$

$$P_2 = D_3 \ D_6 \ D_7 \ D_{10} \ D_{11} \\ 1 \ 0 \ 1 \ 1 \ 0 \Rightarrow 01$$

$$P_4 = D_5 \ D_6 \ D_7 \ D_{12} \\ 0 \ 0 \ 1 \ 1 \Rightarrow 00$$

$$P_8 = D_9 \ D_{10} \ D_{11} \ D_{12} \\ 1 \ 1 \ 0 \ 1 \Rightarrow 0$$

error is at 3rd position

~~101101001110~~ H.C = 10111101

Q. Obtain the 15 bit H.C. for the 11 bit data word
 11001001010.

1	1	0	0	1	0	0	1	1	0	1	0	0	0	0
D_{15}	D_{14}	D_{13}	D_{12}	D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1

$$P_1 = D_3 \ D_5 \ D_7 \ D_9 \ D_{11} \ D_{13} \ D_{15} \\ 0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 0 \Rightarrow 0$$

$$P_2 = D_3 \ D_6 \ D_7 \ D_{10} \ D_{11} \ D_{14} \ D_{15}$$

$$0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 1 \Rightarrow 0$$

$$P_4 = D_5 \ D_6 \ D_7 \ D_{12} \ D_{13} \ D_{14} \ D_{15}$$

$$1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1 \Rightarrow 0$$

$$P_8 = D_9 \ D_{10} \ D_{11} \ D_{12} \ D_{13} \ D_{14} \ D_{15}$$

$$0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1 \Rightarrow 1$$

~~Error~~ is at

$$H.C. = 110010001011011 \text{ at odd}$$

$$H.C. = 110010011010000 \text{ at even}$$

Logic gates

- (1) Basic gates (AND, OR, NOT)
- (2) Universal gates (NOR, NAND)
- (3) Exclusive / special gates (XOR, XNOR)

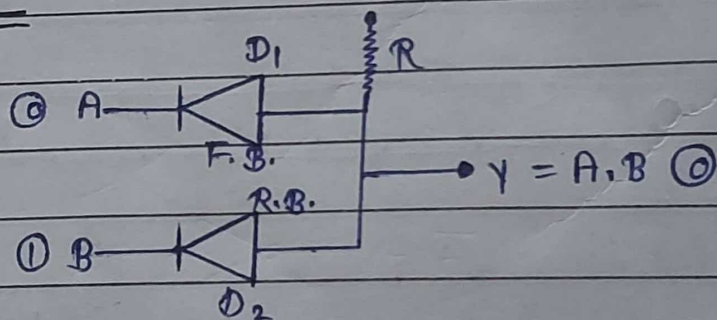
① AND Gate :



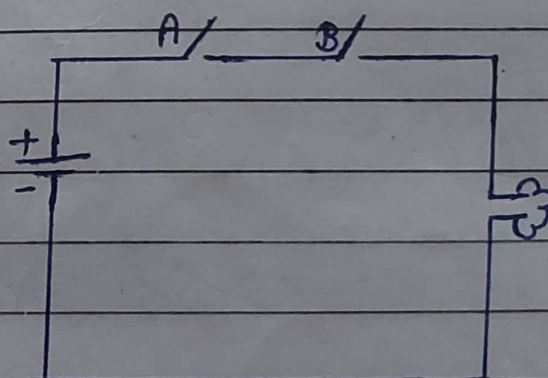
Truth table :

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Circuit diagram :

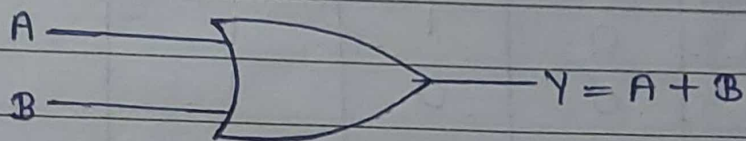


Switching circuit :



A	B	C	$Y = A \cdot B \cdot C$
0	0	0	0
0	1	0	0
0	0	1	0
1	0	0	0
1	0	1	0
1	1	0	0
0	1	1	0
1	1	1	1

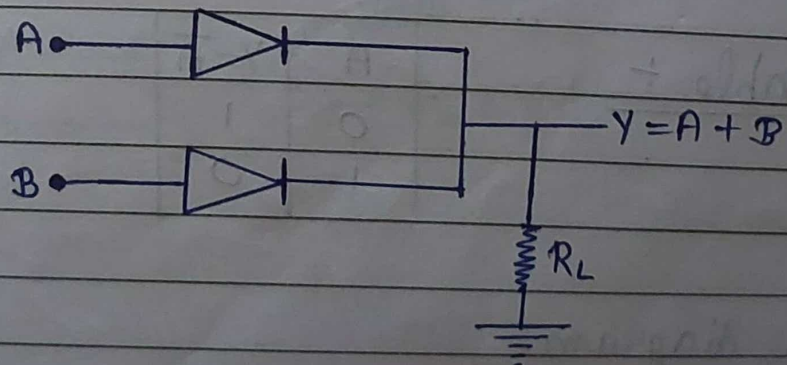
2. OR Gate ÷



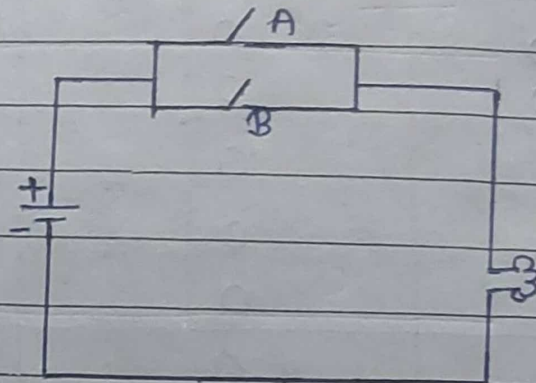
Truth table ÷

A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

Circuit diagram ÷



Switching circuit ÷



A	B	C	$Y = A + B + C$
0	0	0	0
0	1	0	1
0	0	1	1
1	0	0	1
1	0	1	1
0	1	1	1
1	1	0	1
1	1	1	1

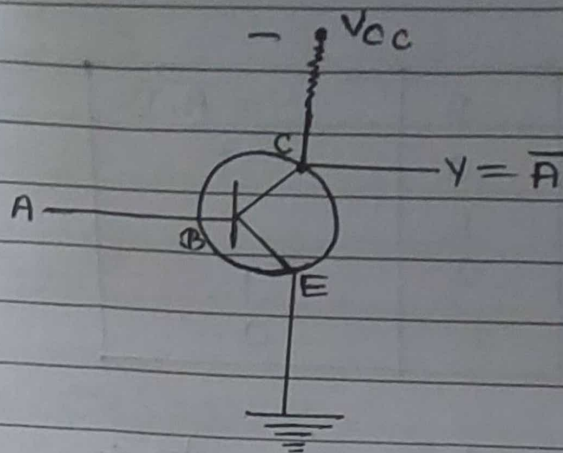
② NOT Gate ÷



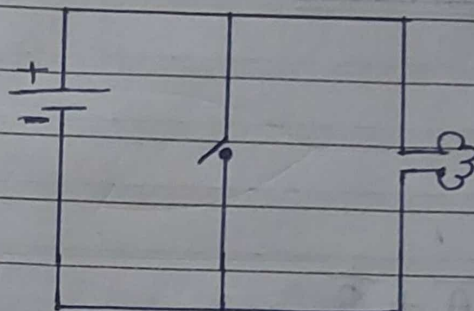
Truth table ÷

A	$Y = \bar{A}$
0	1
1	0

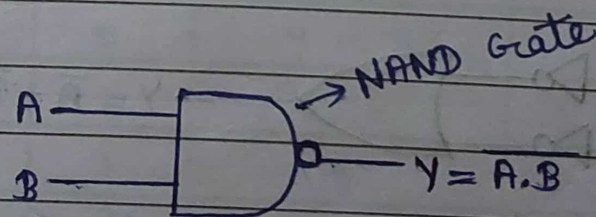
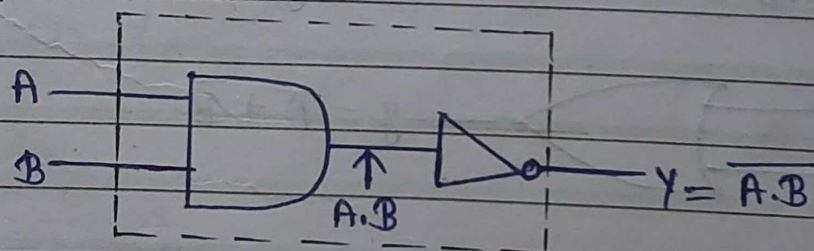
Circuit diagram ÷

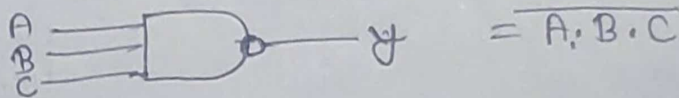


Switching circuit ÷



NAND Gate ÷





Truth table:

A	B	$Y = \overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

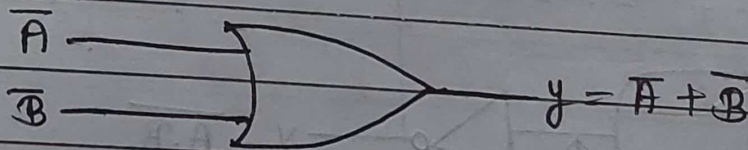
NAND gate as a bubble OR gate:

DeMorgan's theorem:

① $\overline{A+B} = \overline{A} . \overline{B}$

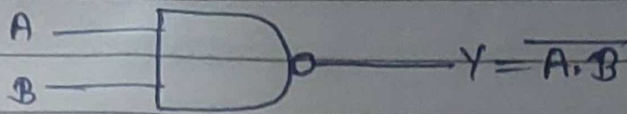
② $\overline{A.B} = \overline{A} + \overline{B}$

$$\begin{array}{ccc} \overline{A.B} & = & \overline{A} + \overline{B} \\ \downarrow & & \downarrow \\ \text{O/P of NAND} & & \text{O/P of NAND gate} \end{array}$$

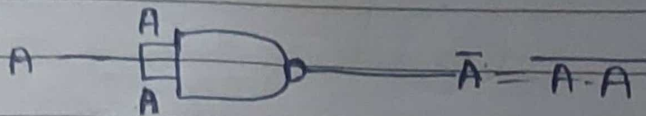


② NAND gate as a universal gate:

a) NAND gate as a NOT gate:

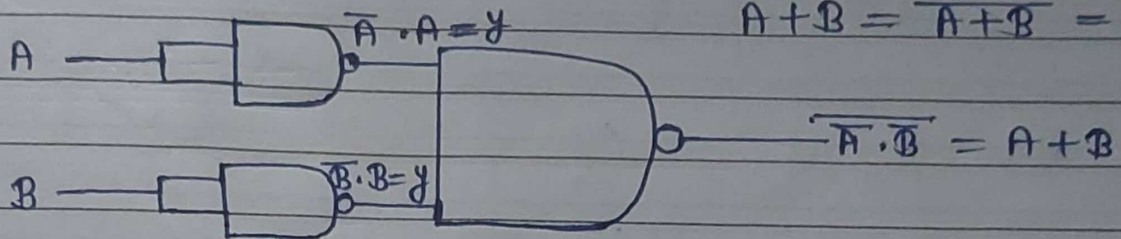


Truth table ÷

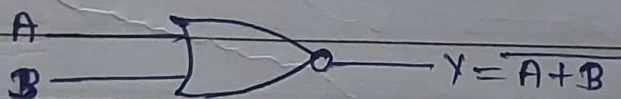
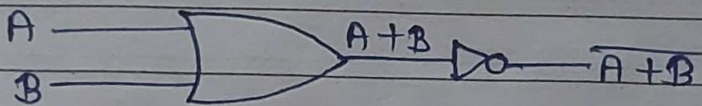


A	$Y = \bar{A}$
0	1
1	0

b) NAND gate as a OR gate ÷

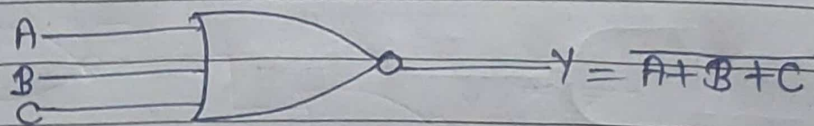


NOR gate ÷



Truth table ÷

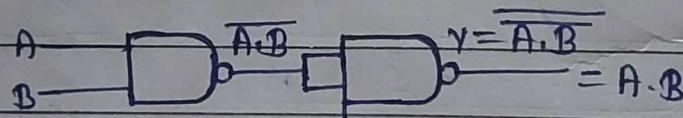
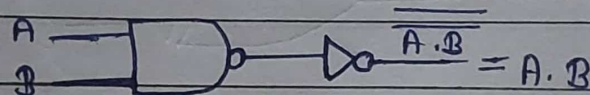
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0



Truth table ÷

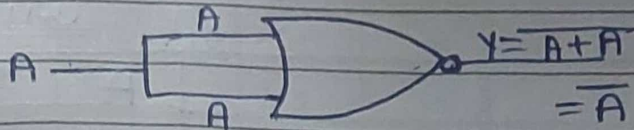
A	B	C	$Y = \overline{A+B+C}$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

(c) NAND gate as a AND gate ÷

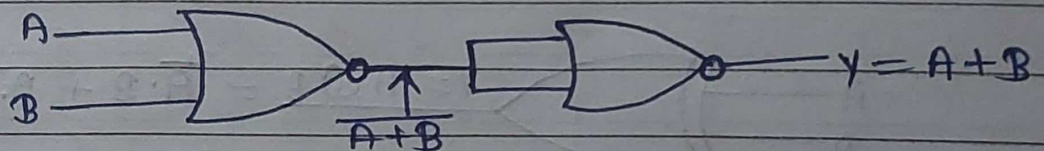


Truth table ÷

A	B	$Y = \overline{\overline{A.B}} = A.B$
0	0	0
0	1	0
1	0	0
1	1	1

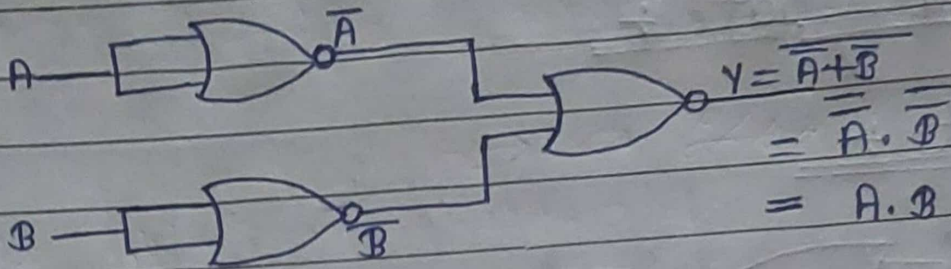
NOR as a universal gate ÷(i) NOR as NOT gate ÷Truth table ÷

A	$Y = \bar{A}$
0	1
1	0

(2) NOR as OR gate ÷Truth table ÷

A	B	$Y = A+B$
0	0	0
0	1	1
1	0	1
1	1	1

(3) NOR as AND gate ÷



Truth table

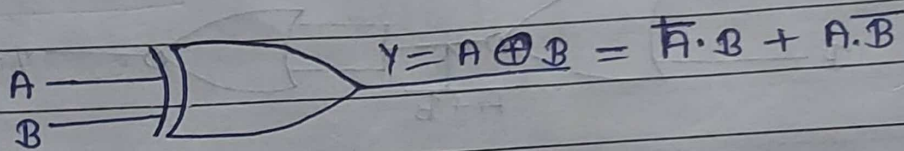
A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

De Morgan's theorem:

- ① $\overline{A+B} = \overline{A} \cdot \overline{B}$
- ② $\overline{A \cdot B} = \overline{A} + \overline{B}$

Special purpose or exclusive gates:

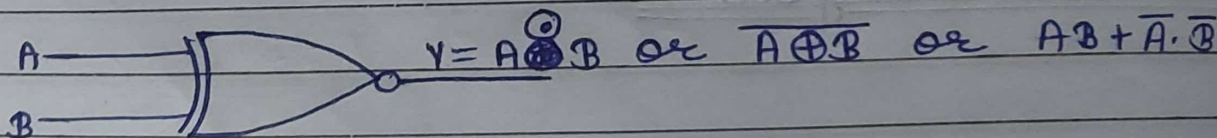
(i) XOR:



Truth table:

A	B	$Y = \overline{A}B + A\overline{B} = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

(2) XNOR:



Truth table :-

A	B	$Y = A \oplus B$
0	0	1
0	1	0
1	0	0
1	1	1

A	B	C	$Y = A \oplus B$ (XOR)	$Y = A \oplus B$ (XNOR)
0	0	0	1	0
0	1	0	0	1
0	0	1	1	0
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
0	1	1	0	0
1	1	1	1	0

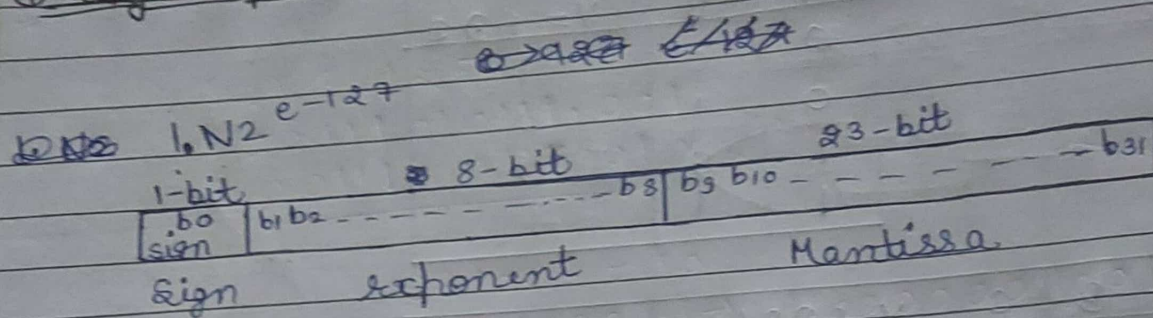
Fixed point and floating point representation :-

Floating point representation :-

Floating point number IEEE 754 has given the format.

By-Single precision format.
double precision format

① Single precision format (32 bit)



Q. Represent 52.21875 in IEEE 754 32 bit floating point format.

$$52.21875$$

$$\Rightarrow 110100.00111$$

$$1.101000011 \times 2^5$$

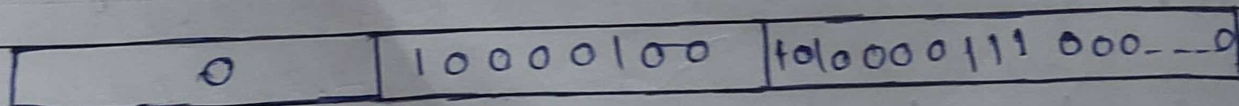
N
(Mantissa)

Mantissa $\rightarrow 101000011$

$$2^{e-127}$$

$$= 2^{e-127} = 5$$

$$= e = 127 + 5 \Rightarrow e = 132 \Rightarrow 10000100$$



Q. Represent (1259.125)

1259.125

1.0011101011.0001

1.0011101011.0001 $\times 2^{10}$

N (Mantissa)

$$e - 127 = 10$$

$$e = 137 \Rightarrow 10001001$$

0 10001001 00111010110001000...0

Q. -307.1875

1.00110011.001

1.00110011.001 $\times 2^{18}$

Mantissa

$$e - 127 = 18$$

$$e = 135 \Rightarrow 10001001$$

1 10001001 00110011001...0

① Double precision format (64 bit) :

1.N2 $e - 1023$

1 bit

11 bit

52 bit



Q. Represent (1259.125) in IEEE double precision format..

1259.125

100111010111.001

1.00111010111001 $\times 2^{10}$

$$e - 1023 = 10$$

$$e = 1033 \Rightarrow 10000001001$$

0 | 10000001001 | 0011101011100100

Unit-2

Boolean algebra:

It is nothing but the mathematics of digital system.

LAWS -

(i) Associate law:

$$(a) A + (B + C) = (A + B) + C$$

$$(b) A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

(2) Commutative law:

$$A + B = B + A$$

$$A \cdot B = B \cdot A$$

(3) Distributive law: